

INTERNSHIP REPORT

Data Cleaning & Reporting Automation using Python and Power BI

Introduction

The objective of this internship project was to automate data cleaning and reporting workflows using Python and Power BI. The project focused on preprocessing a messy employee dataset by handling missing values, removing duplicates, correcting inconsistent data, and generating automated analytical reports.

Data cleaning and reporting automation are important processes in data analytics because clean and well-structured data improves reporting accuracy, business analysis, and decision-making efficiency.

This project was implemented using Python for automation and Power BI for dashboard visualization.

Objective

The main objectives of this project were:

- To automate data preprocessing workflows
- To handle missing values and inconsistent data
- To remove duplicate records
- To standardize and validate employee data
- To generate automated reports and visual summaries
- To create an interactive Power BI dashboard for reporting analysis

Tools and Technologies Used

Tool/Technology	Purpose
Python	Data cleaning and automation
Pandas	Data preprocessing and analysis
Matplotlib	Data visualization
Power BI	Interactive dashboard creation
CSV Dataset	Employee data source

Dataset Description

The dataset used for this project was a messy employee dataset containing employee information and organizational details. The dataset included the following fields:

- Employee_ID
- First_Name
- Last_Name
- Age
- Department_Region
- Status
- Join_Date
- Salary
- Email
- Phone
- Performance_Score
- Remote_Work

The dataset intentionally contained missing values and inconsistent records, making it suitable for automation and preprocessing analysis.

Data Preprocessing

The dataset was imported into Python using Pandas and analyzed for inconsistencies and missing values.

The following preprocessing tasks were completed:

- Checked for missing values in all columns
- Removed duplicate records
- Standardized text formatting for object columns
- Filled missing numerical values using mean values
- Filled missing categorical values using default placeholders
- Validated employee age and salary values
- Removed invalid records and inconsistent entries

The preprocessing workflow was fully automated using Python scripting.

Automation Workflow

An automated preprocessing and reporting workflow was developed using Python.

The following automation tasks were implemented:

- Automatic missing value handling
- Duplicate detection and removal
- Data validation and standardization
- Automated data quality calculation
- Automated summary report generation
- Automatic export of cleaned datasets
- Automatic generation of visual analysis charts

The automation process significantly reduced manual preprocessing effort and improved reporting efficiency.

Automated Reporting

An automated summary report was generated containing:

- Total number of records
- Duplicate records removed
- Remaining missing values
- Data quality score

The cleaned dataset and automated reports were exported as CSV files for further analysis and visualization.

Generated output files:

- Cleaned_Employee_Data.csv
- Automation_Report.csv

Data Visualization & Dashboard Development

An interactive dashboard was developed using Power BI to visualize preprocessing results and employee analytics.

The dashboard included:

- KPI cards for employee count and data quality score
- Department distribution analysis
- Salary analysis by department
- Performance score analysis
- Remote work distribution
- Interactive slicers and filters for dynamic reporting

The dashboard was designed to provide clear visualization of automation results and employee-related insights.

Business Insights

The following insights were identified during the analysis:

- Automated preprocessing improved overall data quality
- Missing values and inconsistent records were successfully handled
- Clean datasets improve reporting accuracy and business analysis
- Automation workflows reduce manual preprocessing effort
- Department and salary analysis provide better organizational insights

Learning Outcomes

Through this internship project, I gained practical knowledge in:

- Data preprocessing and cleaning
- Automation workflows using Python
- Missing value handling techniques
- Duplicate detection and removal
- Data validation and standardization
- Automated reporting systems
- Data visualization using Power BI
- Interactive dashboard development

Conclusion

The Data Cleaning & Reporting Automation project was successfully completed using Python and Power BI. The project effectively automated the preprocessing of messy employee datasets and generated analytical reports and visual dashboards.

The automation workflow improved data quality, reduced manual effort, and enhanced reporting efficiency. This project strengthened my understanding of data preprocessing, automation techniques, and business reporting workflows in real-world analytics environments.